IT SERVICE DESK MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "IT SERVICE DESK MANAGEMENT SYSTEM" is the bonafide work of Aakif Umar Z (2116241501002), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, AAKIF UMAR Z (2116241501002), hereby declare that the project report titled “IT SERVICE DESK MANAGEMENT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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Date:

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

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ABSTRACT

IT support requests in many organisations are raised through telephone calls, e-mails and informal messages, which makes them difficult to categorise, prioritise, assign and follow up. Requests can be lost or duplicated, users do not know who is working on their problem, and the support team has no consolidated view of open tickets, resolved tickets or the workload of each agent.

This project presents IT SERVICE DESK MANAGEMENT SYSTEM, a IT ticket support system developed on the Zoho Creator low-code platform. The application is built around five forms: Users, Support Agents, Categories, Tickets and Ticket Comments. A user submits a ticket with a title, description, category, priority and an optional attachment; an administrator assigns it to a support agent; and the agent moves it through the statuses Open, In Progress, Resolved and Closed while adding comments. The creation date and the resolution date are recorded for every ticket. Reports are provided for tickets, agents, categories and users, and a dashboard presents key performance indicators and charts for ticket status, priority, category and agent workload. Role-based access separates Users, Support Agents and Admins wherever the platform supports it, and short Deluge scripts validate ticket data.

Functional test cases are defined for form validation, ticket submission and assignment, status tracking, comments, reports and role-based access, and a format for recording user acceptance feedback is provided. The project shows that a maintainable, browser- based and mobile-accessible helpdesk can be obtained without building a conventional web application, giving every request a traceable record from submission to resolution. E-mail notification, service-level timers and integrations with external tools are identified as future enhancements and are outside the scope of this version.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; IT Service Management; Helpdesk Automation; Role-Based Access Control*

PROJECT AT A GLANCE

Attribute	Details
Project Title	IT SERVICE DESK MANAGEMENT SYSTEM
Application Domain	IT Service Management / Helpdesk Automation
Student Name & Register No.	AAKIF UMAR Z - 2116241501002
Zoho Creator Edition	Education – Trial version
Application Name (Link Name)	IT SERVICE DESK MANAGEMENT SYSTEM
Published / Live URL	https://creatorapp.zoho.in/aakif18cric/it-service-desk-management-system1#Section_1
Demo Credentials (per role)	**Admin:** Not configured | **Support Agent:** Not configured | **User:** Not configured
Number of Forms / Subforms	5 / 0
Number of Reports (by type)	List: 5 | Kanban: 0 | Calendar: 0 | Pivot/Chart: 0 | Others: 0
Number of Pages / Dashboards	1-Helpdesk Dashboard with KPIs and charts
Number of Workflows	Form: 3 | Schedule: 0 | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	3 – DS-01, DS-02, DS-03
Integrations	The application supports built-in file attachment uploads for support tickets. E-mail notifications and Zoho Analytics integration are planned as future enhancements and have not yet been implemented.
Roles & Profiles Defined	Admin, Support Agent, User
Environments Used	Single environment – application developed and published directly
Date of Production Deployment	

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR	Functional Requirement
IT	Information Technology
ITIL	Information Technology Infrastructure Library
ITSM	IT Service Management
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Every organization that depends on computers, networks and business software needs an IT support function that receives problems from its users and resolves them. Typical requests concern hardware faults, software installation, network connectivity, account and login problems, e-mail, printers and other issues. Each request must be recorded, classified, given a priority, assigned to a suitable support agent, followed until it is resolved, and closed with a record of when it was raised and when it was resolved. This discipline is part of IT service management, for which the ITIL framework describes the service desk and incident-handling practices [8].

1.2 Problem Statement

Managing IT support requests becomes difficult when they are scattered across telephone calls, e-mails and spreadsheets. Requests are not classified or prioritised in a consistent way, assignment depends on the administrator remembering who is available, users cannot see the progress of their request, and there is no reliable record of when a problem was raised, who handled it and when it was resolved. A centralised, structured and access-controlled ticketing application is therefore required to simplify request submission, assignment, tracking and communication between users and support agents. The specific problems are:

- Requests arrive through several channels, so there is no single place that holds every open issue.
- Issues are not categorised or prioritised consistently, so urgent faults may wait behind minor ones.
- Assignment is manual, and the workload and availability of each support agent are not visible.
- Users cannot see the status of a request, and updates and screenshots are exchanged outside the request record.
- Creation and resolution dates are not recorded systematically, so reports on tickets, categories and agents are hard to produce.

1.3 Objectives

- To design a normalised data model and forms for users, support agents, categories, tickets and ticket comments.
- To allow users to submit categorised, prioritised tickets with supporting attachments and to track them through the statuses Open, In Progress, Resolved and Closed.
- To support assignment of tickets to support agents and communication through comments on each ticket.
- To provide reports for tickets, agents, categories and users, and a dashboard with KPIs and charts for status, priority, category and agent workload.
- To restrict access through role-based permissions for Users, Support Agents and Admins wherever the platform supports it.
- To validate ticket data with short Deluge scripts, and to evaluate the application through test cases and user feedback.

1.4 Scope of the Project

The application contains five forms: Users, Support Agents, Categories, Tickets and Ticket Comments. It supports submission of new IT support requests, categorisation under Hardware, Software, Network, Account / Login, Email, Printer and Other, assignment of priorities, assignment of tickets to support agents, tracking of the status Open, In Progress, Resolved and Closed, comments and updates on tickets, attachment of screenshots and other files, and recording of creation and resolution dates. Reports are provided for tickets, agents, categories and users, and a dashboard presents KPIs and charts for ticket status, priority, category and agent workload. Three kinds of user are considered – Users, Support Agents and Admins – and role-based access is applied wherever the platform supports it.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because the problem consists mainly of structured records – users, agents, categories, tickets and comments – that are naturally represented as forms with related fields. Low-code platforms provide visual tools for building forms, reports and workflows, and allow developers to concentrate on business requirements rather than on infrastructure and repetitive code [1], [3], [4]. Zoho Creator also provides built-in hosting, user management, role-based access, file-upload fields, a mobile-friendly

interface and the Deluge scripting language for logic that cannot be configured visually [2], [5].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing way of working, users report IT problems by telephone, e-mail, messaging groups or in person. A support staff member notes the request in an inbox or a spreadsheet, and the administrator decides who should handle it and informs that person verbally or by e-mail. The agent works on the issue and gives updates through calls and messages, and users who do not hear back ask again for the status.

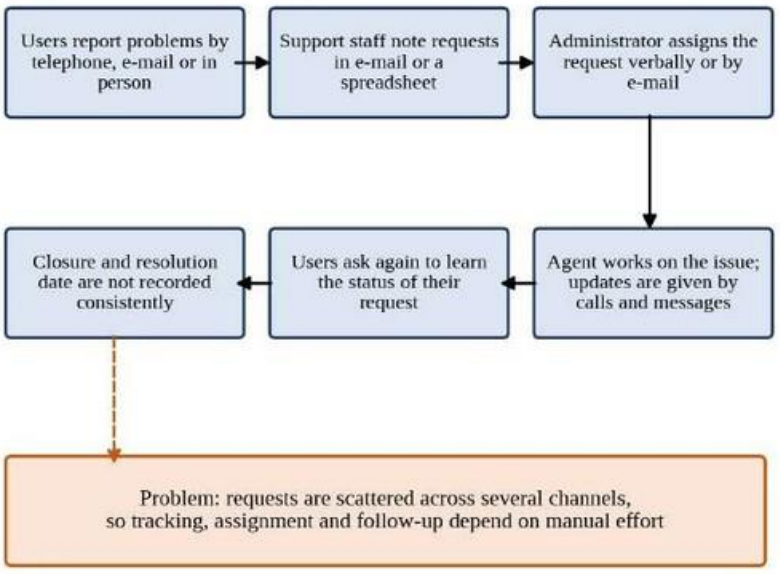


Figure 2.1: As-Is Process Flow of the Existing IT Support Request Handling

2.2 Limitations of the Existing System

- Requests are spread across calls, e-mails and spreadsheets, so there is no single trusted list of open issues.
- Categories and priorities are applied inconsistently, which delays urgent problems.
- Assignment is manual, and the availability and workload of agents are not visible. • Users cannot see progress without contacting the support team, which creates repeated enquiries.
- Screenshots and supporting files are sent separately from the request they explain.

• Creation and resolution dates are not recorded consistently, and there is no access control or traceable history.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Needs separate design and coding of the database, back end, user interface and security; longest for an application with several forms and reports.	Forms, reports and pages are configured visually; the five forms and their lookups can be built quickly, and Deluge is used only for short validation logic.
Skills required	Front-end, back-end, database and deployment skills.	Understanding of data modelling and the platform; basic Deluge scripting.
Hosting & infrastructure	Server, database, domain and deployment pipeline must be arranged and maintained.	Hosting, backup and scaling are provided by the Zoho Creator platform, subject to the selected plan.
Mobile application	A separate responsive front end or native app would have to be developed.	The application can be opened in a mobile browser and in the Zoho Creator mobile app without extra development.
Security & access control	Authentication, authorisation and session handling must be implemented.	Users, roles, permission profiles and record-level criteria are built in and configured, not coded.
Maintenance & change	A change to a field or report may touch several layers of code and need redeployment.	A new field, category or report can usually be changed by the administrator in the builder.
Customisation limits	Almost unlimited design freedom.	Limited by the platform's field types, page builder and Deluge capabilities (see Section 8.4).
Cost	Developer time, hosting and maintenance costs.	Platform subscription according to the plan; the Education edition may be used for learning.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	E. 't Hoen [1]	Low Code Development with Mendix (BPB Publications)	Book that introduces low-code development and how applications are built with visual models, logic and security configuration.	Provides the conceptual basis for choosing and applying a low-code platform for Helpdesk Pro.
2	Zoho Corporation [2]	Zoho Creator Help Documentation (online)	Describes forms, field types, reports, pages, workflows, user management and publishing in Zoho Creator.	Reference for designing the five forms, reports, dashboard and permissions.
3	R. Sanchis, Ó. García-Perales, F. Fraile and R. Poler, 2020 [3]	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Argues that low-code development platforms facilitate and automate the development of software applications that support enterprise needs and digital transformation, and benchmarks existing platforms.	Academic support for using a low-code platform to digitise an organisational process.
4	A. C. Bock and U. Frank, 2021 [4]	Low-Code Platform	Characterises low-code platforms and discusses their features and their implications for application development.	Defines the low-code platform concept used in Chapter 1.
5	Zoho Corporation [5]	Deluge Script Reference Guide (online)	Describes the Deluge language, form events, fetch statements and the sendmail task.	Reference for the validation scripts DS-01 to DS-03.
6	Zoho Corporation [6]	Zoho Creator Learning Centre (online)	Provides tutorials and learning material for building Zoho Creator applications.	Used as a learning resource while developing and testing the application.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
7	D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi and M. Wimmer, 2022 [7]	Low-code development and model-driven engineering: Two sides of the same coin?	Examines the relationship between low-code development and model-driven engineering.	Explains why form and data models are the foundation of a low-code application.
8	AXELOS, 2019 [8]	ITIL Foundation: ITIL 4 Edition	Describes IT service management practices, including the service desk and incident management.	Provides the service-management context for a ticket-based helpdesk.

2.5 Emerging Trends in Low-Code Development

Several trends in low-code development influenced the design of Helpdesk Pro. The first is citizen development: platforms now allow people who are not professional programmers to build and maintain business applications, which suits an IT support team that must keep the application running after the project is complete [3], [4]. The portal therefore uses visual configuration wherever possible and keeps Deluge scripts short.

2.6 Proposed System

The proposed Helpdesk Pro is a centralised IT support ticket system developed with Zoho Creator. It stores users, support agents, categories, tickets and ticket comments in five related forms, so that every request has one record that holds its category, priority, status, assigned agent, attachment, creator and dates.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Admin	IT support administrator who manages the helpdesk.	Maintains users, support agents and categories; assigns tickets to agents; monitors reports and the dashboard.
Support Agent	IT support staff member who resolves tickets.	Views assigned tickets; updates the ticket status; adds comments; records the resolution.
User	Employee or member who needs IT help.	Submits tickets with category, priority and attachments; tracks their own tickets; adds comments.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain user records with name, department, role and status.	Admin	High
FR-02	The system shall maintain support agent records with contact information, department, expertise and availability.	Admin	High
FR-03	The system shall maintain the categories Hardware, Software, Network, Account / Login, Email, Printer and Other.	Admin	Medium
FR-04	The system shall allow a user to submit a ticket with a title, description, category and priority.	User	High
FR-05	The system shall allow screenshots and other supporting files to be attached to a ticket.	User	Medium
FR-06	The system shall allow a ticket to be assigned to a support agent.	Admin	High
FR-07	The system shall track the ticket status as Open, In Progress, Resolved or Closed.	Support Agent, Admin	High
FR-08	The system shall record the ticket creation date and the resolution date and keep them consistent.	Support Agent, Admin	Medium

FR ID	Requirement	Role(s)	Priority
FR-09	The system shall allow comments and updates to be added to a ticket.	User, Support Agent	High
FR-10	The system shall allow a user to view their own tickets and the comments on them.	User	High
FR-11	The system shall allow a support agent to view the tickets assigned to them.	Support Agent	High
FR-12	The system should provide reports for tickets, agents, categories and users.	Admin	Medium
FR-13	The system should provide dashboard KPIs and charts for ticket status, priority, category and agent workload.	Admin, Support Agent	Medium
FR-14	The system shall restrict access and operations according to the configured role permissions.	All users	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	The application shall provide a clear interface through which a user can submit a ticket and find its status within a few clicks.
NFR-02	Performance	Forms and reports shall load within an acceptable time under normal use; reports shall use filters so that only relevant records are loaded.
NFR-03	Security	Users shall access only the data and operations permitted by their role; administrative operations shall be restricted to authorised users.
NFR-04	Maintainability	Categories, field choices and reports shall be maintained in the builder so that changes do not require reprogramming.
NFR-05	Accessibility / Usability on devices	The application shall be usable in supported browsers and on mobile devices through the browser or the Zoho Creator mobile app.
NFR-06	Data integrity	Mandatory fields shall be validated before a ticket is accepted; resolution dates shall not precede creation dates.
NFR-07	Reliability	Records shall remain consistent across related forms; a ticket or comment shall always refer to an existing related record.
NFR-08	Auditability	Important changes shall be traceable through the audit trail and change history available on the platform.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman (used only if an external API is tested)
Other Zoho services	File attachment (built-in upload field); e-mail notification and Zoho Analytics are planned and not configured
Mobile	Zoho Creator mobile app (Android / iOS) or a mobile browser
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

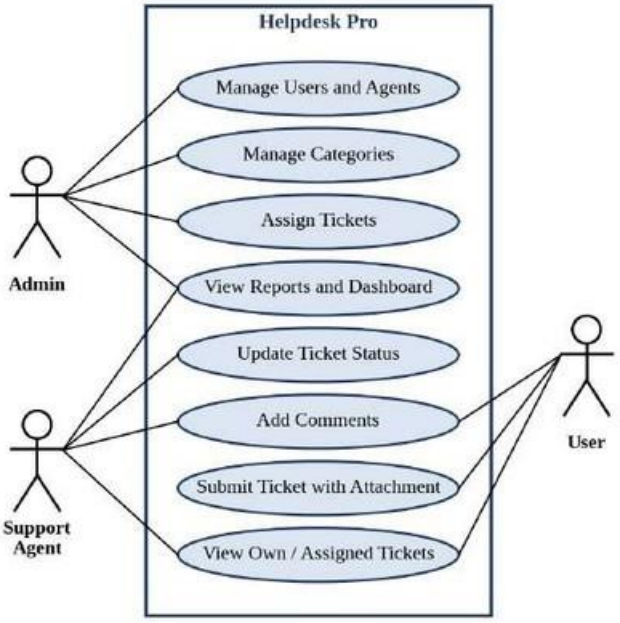


Figure 3.1: Use Case Diagram of Helpdesk Pro

Table 3.5: Use Case Description – Submit Ticket

Use Case ID	UC-01
Use Case Name	Submit Ticket
Actor(s)	User
Pre-condition	The user is signed in with the User role; the categories exist in the Categories form.
Main Flow	1. The user opens the Tickets form. 2. The user enters the title and description and selects the category and priority. 3. The user attaches a screenshot or file, if needed. 4. The user submits the form. 5. The application saves the ticket with the status Open and the creation date.
Alternate Flow	A mandatory field is empty; the application shows a validation message and does not save the ticket.
Post-condition	A new ticket exists with the status Open and is visible to the user and to the Admin.
Related FR	FR-04, FR-05, FR-08, FR-10

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

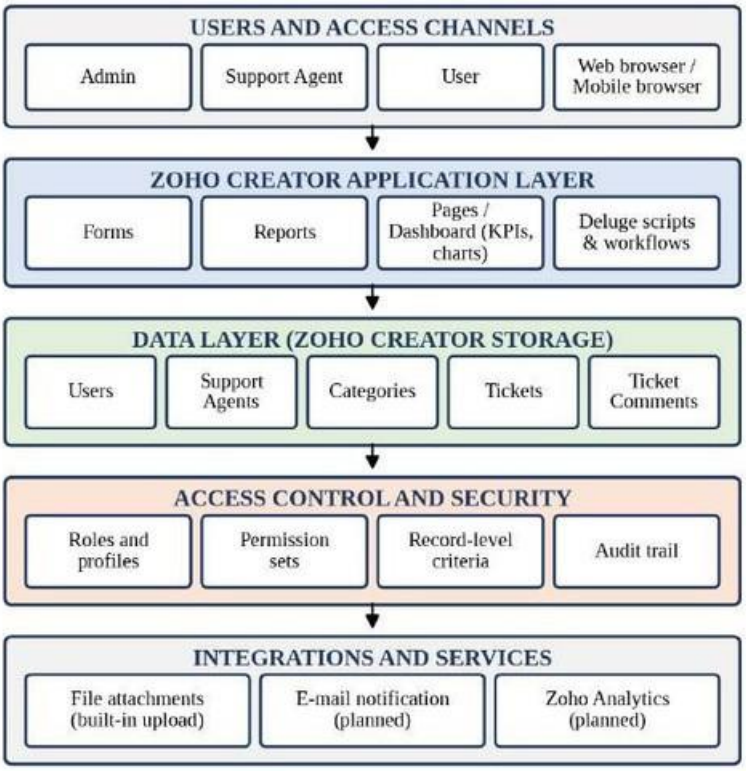


Figure 4.1: System Architecture of the College Bus Transportation Portal

Helpdesk Pro follows a layered architecture on Zoho Creator, shown in Figure 4.1. The topmost layer contains the users and their access channels: the Admin, the Support Agent and the User, who reach the application through a web browser or a mobile browser. The application layer contains the components configured in Zoho Creator: the forms that capture data, the reports that list and filter records, the dashboard page with KPIs and charts, and the Deluge scripts and form workflows.

4.2 Data Model

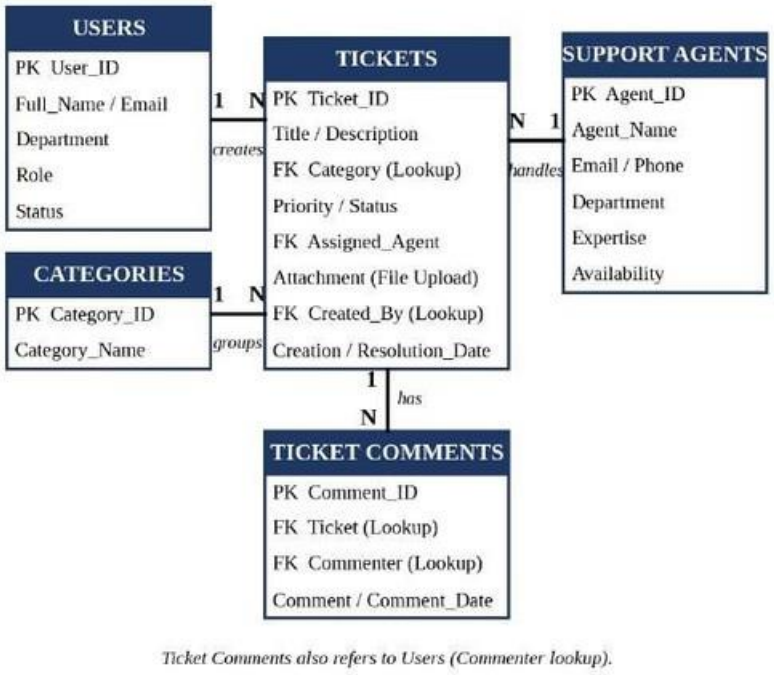


Figure 4.2: Entity–Relationship Diagram of Helpdesk Pro

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Users	User_ID, Full_Name, Email, Department, Role, Status	Tickets, Ticket Comments	One-to-Many	Lookup fields Created_By and Commenter
Support Agents	Agent_ID, Agent_Name, Email, Phone, Department, Expertise, Availability	Tickets	One-to-Many	Lookup field Assigned_Agent in Tickets
Categories	Category_ID, Category_Name	Tickets	One-to-Many	Lookup field Category in Tickets
Tickets	Ticket_ID, Title, Description, Category, Priority, Status, Assigned_Agent, Attachment, Created_By, Creation_Date, Resolution_Date	Users, Categories, Support Agents, Ticket Comments	Many-to-One (three lookups); One-to-Many with comments	Lookup fields Category, Assigned_Agent, Created_By
Ticket Comments	Comment_ID, Ticket, Commenter, Comment, Comment_Date	Tickets, Users	Many-to-One	Lookup fields Ticket and Commenter

Tickets is the central transactional form. Users, Support Agents and Categories hold master data that changes rarely, and each ticket refers to them through lookup fields, so a person, an agent or a category is stored once and reused. One ticket can have many comments, and each comment refers to its ticket. No subforms are used. Assigned_Agent is optional so that a new ticket can exist before it is assigned.

4.3 Form Design

Table 4.2: Field Specification – Users Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
User ID	User_ID	Auto Number	—	Prefix USR-	System generated
Full Name	Full_Name	Single Line	Yes	—	Name of the user
Email	Email	Email	Yes	Valid e-mail address	Contact address
Department	Department	Single Line	Yes	—	User's department
Role	Role	Drop Down	Yes	—	User / Support Agent / Admin
Status	Status	Drop Down	Yes	Default: Active	Active Under Maintenance/ Inactive

Table 4.3: Field Specification – Support Agents Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Agent ID	Agent_ID	Auto Number	—	Prefix AGT-	System generated
Agent Name	Agent_Name	Single Line	Yes	—	Name of the agent
Email	Email	Email	Yes	Valid e-mail address	Contact e-mail
Phone	Phone	Phone	No	Valid phone number	Contact number
Department	Department	Single Line	No	—	Agent's department
Expertise	Expertise	Single Line	No	—	Area of expertise

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Availability	Availability	Drop Down	Yes	—	Available, Busy, On Leave

Table 4.4: Field Specification – Categories Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Category ID	Category_ID	Auto Number	—	Prefix CAT-	System generated
Category Name	Category_Name	Single Line	Yes	No duplicate values	Hardware, Software, Network, Account / Login, Email, Printer, Other

Table 4.5: Field Specification – Tickets Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Ticket ID	Ticket_ID	Auto Number	—	Prefix TKT-	System generated
Title	Title	Single Line	Yes	—	Short summary of the issue
Description	Description	Multi Line	Yes	—	Details of the problem
Category	Category	Lookup	Yes	Lookup to Categories	Type of issue
Priority	Priority	Drop Down	Yes	—	High, Low, Medium, Critical
Status	Status	Drop Down	Yes	Default: Open	Open / In Progress / Resolved / Closed
Assigned Agent	Assigned_Agent	Lookup	No	Lookup to Support Agents	Empty until assigned
Attachment	Attachment	File Upload	No	—	Screenshot or supporting file
Created By	Created_By	Lookup	Yes	Lookup to Users	Creator of the ticket
Creation Date	Creation_Date	Date	Yes	Default: today	Date the ticket is raised

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Resolution Date	Resolution_Date	Date	No	DS-01, DS-02	Date the ticket is resolved

Table 4.6: Field Specification – Ticket Comments Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Comment ID	Comment_ID	Auto Number	—	Prefix CMT-	System generated
Ticket	Ticket	Lookup	Yes	Lookup to Tickets (DS-03)	Related ticket
Commenter	Commenter	Lookup	Yes	Lookup to Users	Author of the comment
Comment	Comment	Multi Line	Yes	—	Comment or update text
Comment Date	Comment_Date	Date	Yes	Default: today	Date of the comment

Nine distinct Zoho Creator field types are used across the five forms: Auto Number, Single Line, Multi Line, Email, Phone, Drop Down, Lookup, File Upload and Date. The Tickets form includes Priority options (Low, Medium, High, Critical) and Status options (Open, In Progress, Resolved, Closed). The Support Agents form includes Availability options (Available, Busy, On Leave).

4.4 Process and Workflow Design

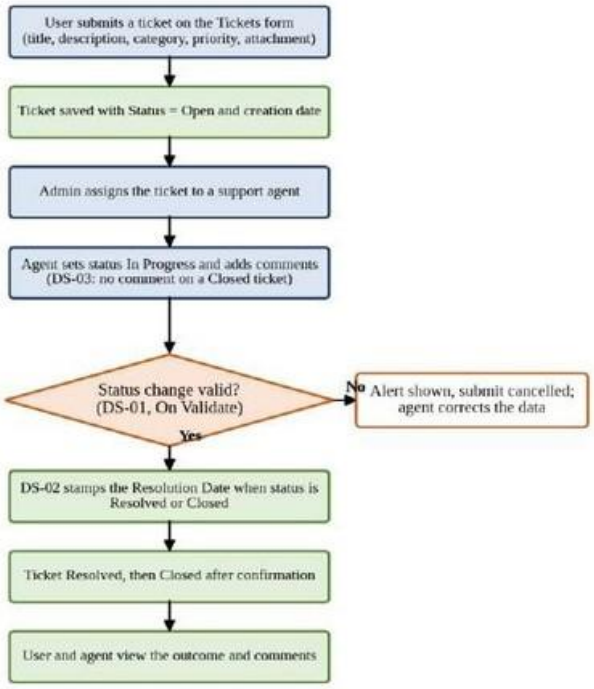


Figure 4.3: Workflow / Process Flow for Ticket Handling

Figure 4.3 shows the ticket life cycle. A user submits a ticket on the Tickets form; it is saved with the status Open and the creation date. The Admin assigns it to a support agent, who changes the status to In Progress and adds comments; the script DS-03 refuses a comment on a Closed ticket. When the status is changed to Resolved or Closed, the validation script DS-01 checks the dates and cancels the submission with an alert if they are inconsistent, and DS-02 stamps the resolution date. The user and the agent can then read the outcome and the comments on the ticket.

4.5 User Interface and Navigation Design

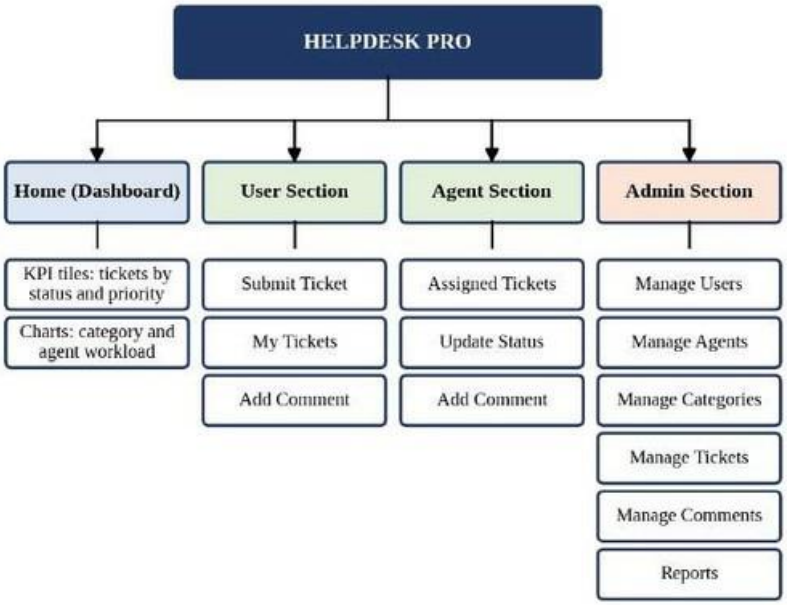


Figure 4.4: Navigation Structure of Helpdesk Pro

The portal has four groups of menus. The Home page is a dashboard with KPI tiles and charts. The User section contains Submit Ticket, My Tickets and Add Comment. The Agent section contains Assigned Tickets, Update Status and Add Comment. The Admin section contains Manage Users, Manage Agents, Manage Categories, Manage Tickets, Manage Comments and Reports. Sections are shown to a user according to the role assigned to that user, wherever the platform supports it.

4.6 Access Control Design

Table 4.7: Role–Permission Matrix

Component	Admin	SupportAgent	User
Form: Users	A E V D	V	V (own)
Form: Support Agents	A E V D	V	—
Form: Categories	A E V D	V	V
Form: Tickets	A E V D	E V (assigned)	A V (own)
Form: Ticket Comments	A E V D	A V (assigned)	A V (own tickets)
Reports: Tickets, Agents, Categories, Users	V	V (assigned)	V (own)
Page: Helpdesk Dashboard	V	V	—
Administrative settings	Full	—	—

A = Add, E = Edit,V=View, D = Delete, — = No access.he Helpdesk Pro application defines three roles:Admin, Support Agent, and User. The Admin manages forms, reports, categories,agents, tickets, and administrative settings. Support Agents view

and update assigned tickets and add comments to support requests. Users create tickets, view their own tickets, and add comments. Access permissions are intended to protect ticket information and ensure that each role can perform its assigned responsibilities.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in Zoho Creator from scratch in the workspace of the student's Zoho account and was named Helpdesk Pro. The first step was to identify the entities of the problem – users, support agents, categories, tickets and ticket comments – and to create one form for each of them, as designed in Chapter 4. Zoho Creator generates a default list report for every form, and these reports were refined later.

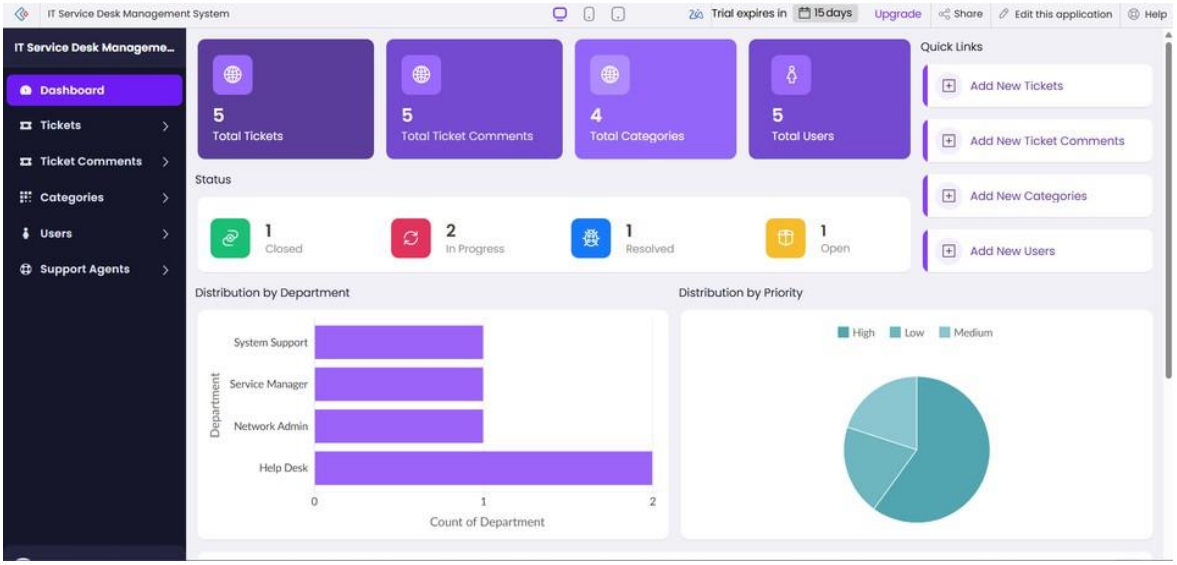


Figure 5.1: Helpdesk – Application Home and Navigation Sections

5.2 Forms and Data Capture

The five forms were built in the Zoho Creator form builder with the field types listed in Tables 4.2 to 4.6. Mandatory fields were marked as required, drop-down fields were given fixed choices and default values, the Category, Assigned Agent, Created By, Ticket and Commenter fields were created as lookups, and the Attachment field uses the File Upload type. The descriptions below explain the rules that make the data reliable.

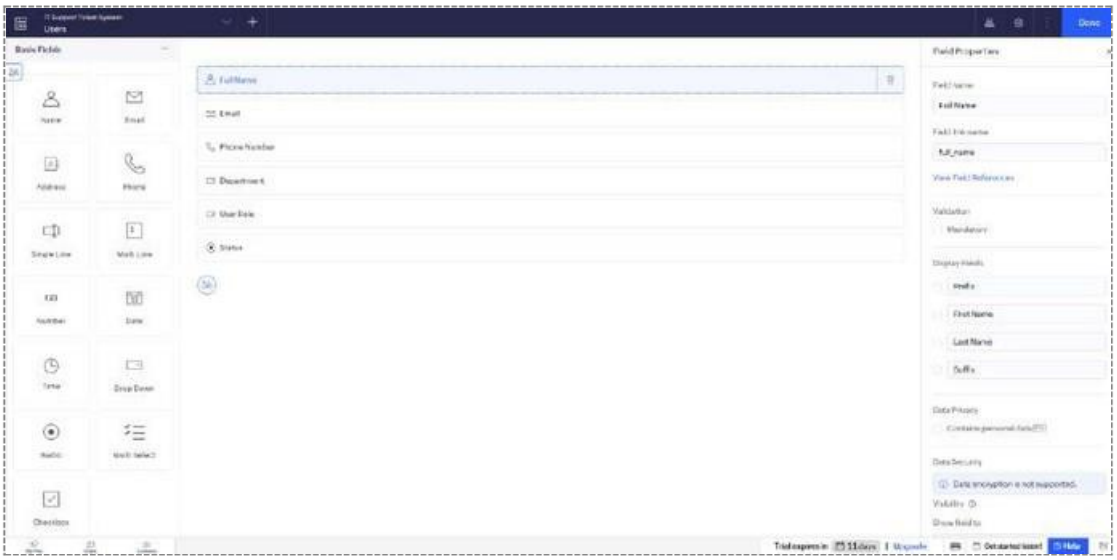


Figure 5.2: Users Form in the Form Builder

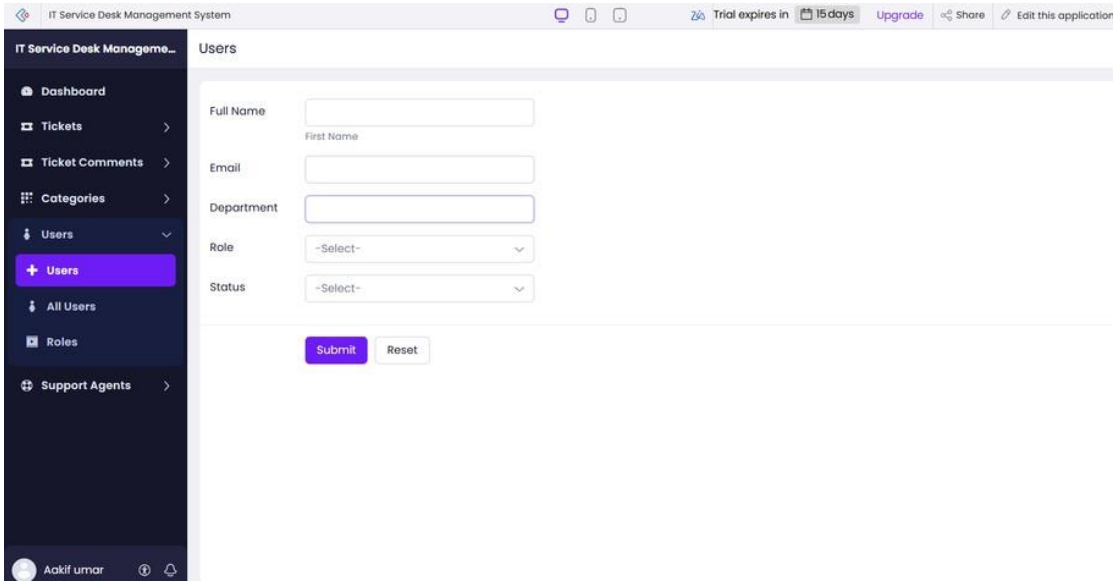


Figure 5.3: Users Form in Live Mode

Support Agents form. This form stores the contact information of each support agent (e-mail and phone), the department, the area of expertise and the availability. The Admin uses it to choose a suitable agent when assigning a ticket.

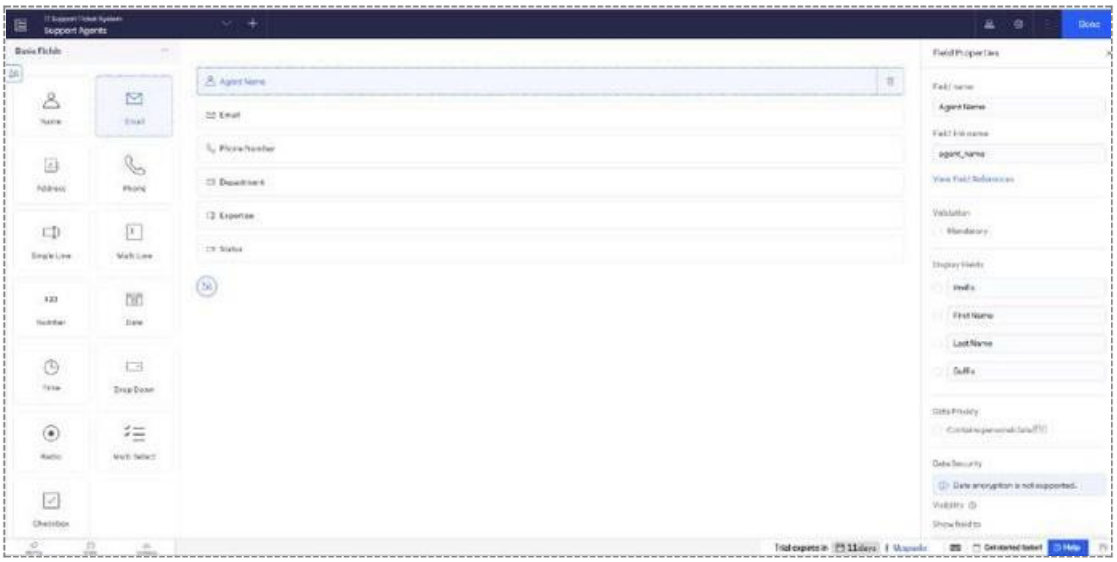


Figure 5.4: Support Agents Form in the Form Builder

Categories and Tickets forms. The Categories form holds the issue categories Hardware, Software, Network, Account / Login, Email, Printer and Other. The Tickets form is the central form: it stores the title, description, category, priority, status, assigned agent, attachment, creator, creation date and resolution date. New tickets default to the status Open and the current date, so a request is never saved without a status. The scripts DS-01 and DS-02 are attached to this form (Section 5.5).



Figure 5.5: Tickets Form in the Form Builder

Ticket Comments form. The Ticket Comments form stores a comment, the ticket it belongs to, the commenter and the comment date. Because comments are separate records, the full conversation of a ticket can be listed in date order, and the script DS-03 checks the status of the related ticket before a comment is accepted.

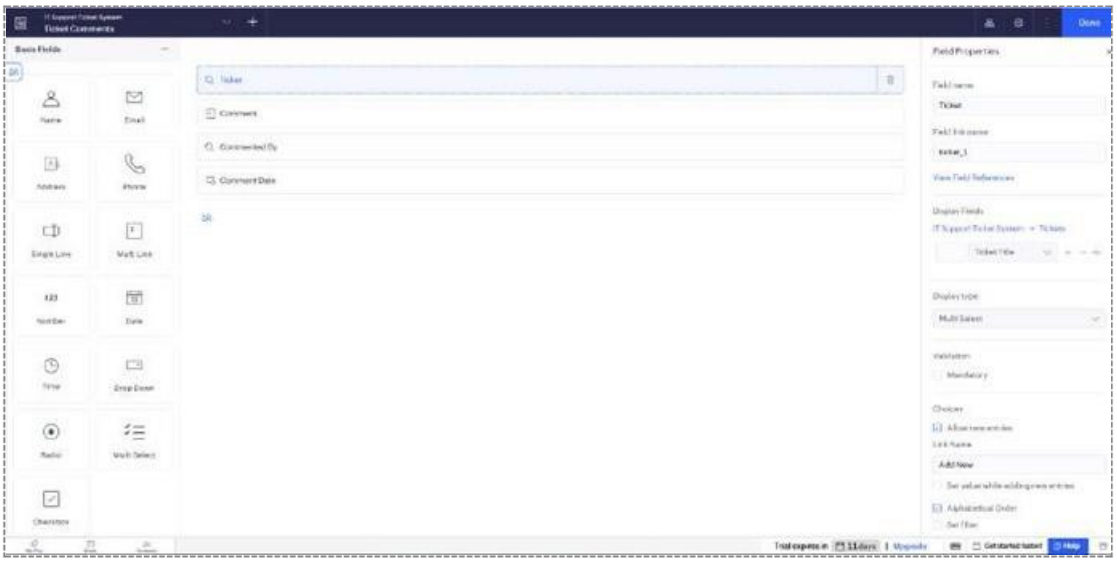


Figure 5.6: Ticket Comments Form in the Form Builder

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Tickets Report	List	Tickets	Sorted by Creation Date; filters on Status, Priority and Category	Edit by Admin and assigned agent	Admin, Support Agent, User (own)
Support Agents Report	List	Support Agents	Sorted by Agent Name; shows Availability	None (view only for agents)	Admin, Support Agent
Categories Report	List	Categories	Sorted by Category Name	None	Admin
Users Report	List	Users	Grouped by Role; filter on Status	Edit / Delete by Admin	Admin

Four list reports were created, one for tickets, agents, categories and users. Each report shows only the columns that users need, and quick search and column filters let a user find a ticket by status, priority or category. Where the platform supports it, record-level criteria limit a User to their own tickets and a Support Agent to the tickets assigned to them.

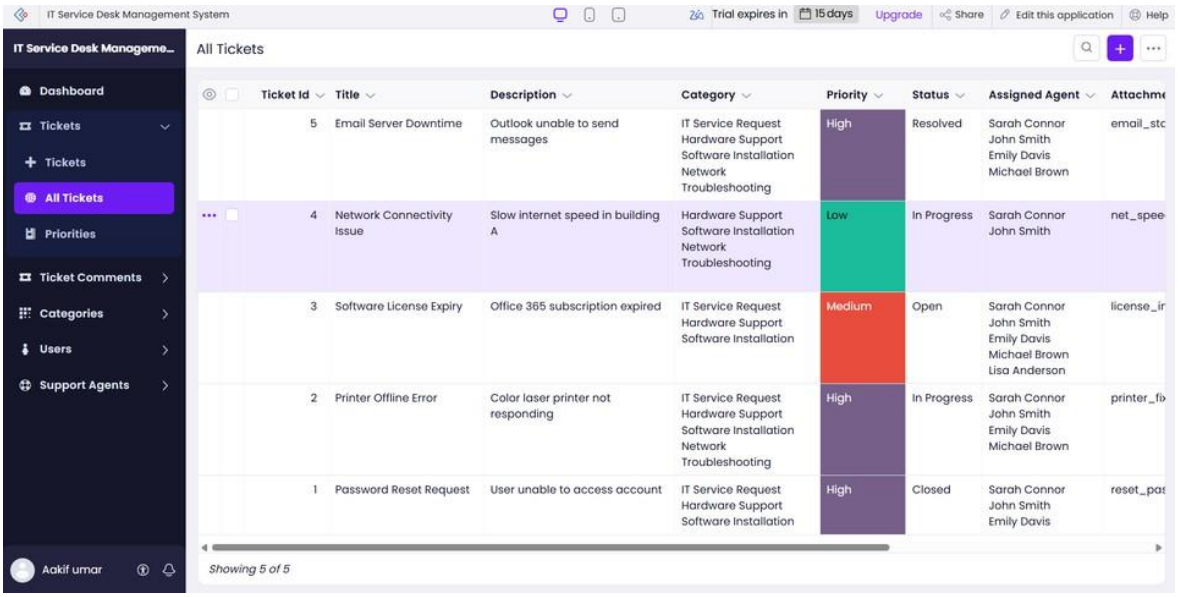


Figure 5.7: Tickets Report

5.4 Pages and Dashboards

A dashboard page named Helpdesk Dashboard was built with the Zoho Creator UI Builder. It presents key performance indicators and charts for ticket status, priority, category and agent workload, so that the Admin can see at a glance how many tickets are Open, In Progress, Resolved or Closed, which categories generate most requests and how the tickets are distributed among the agents.

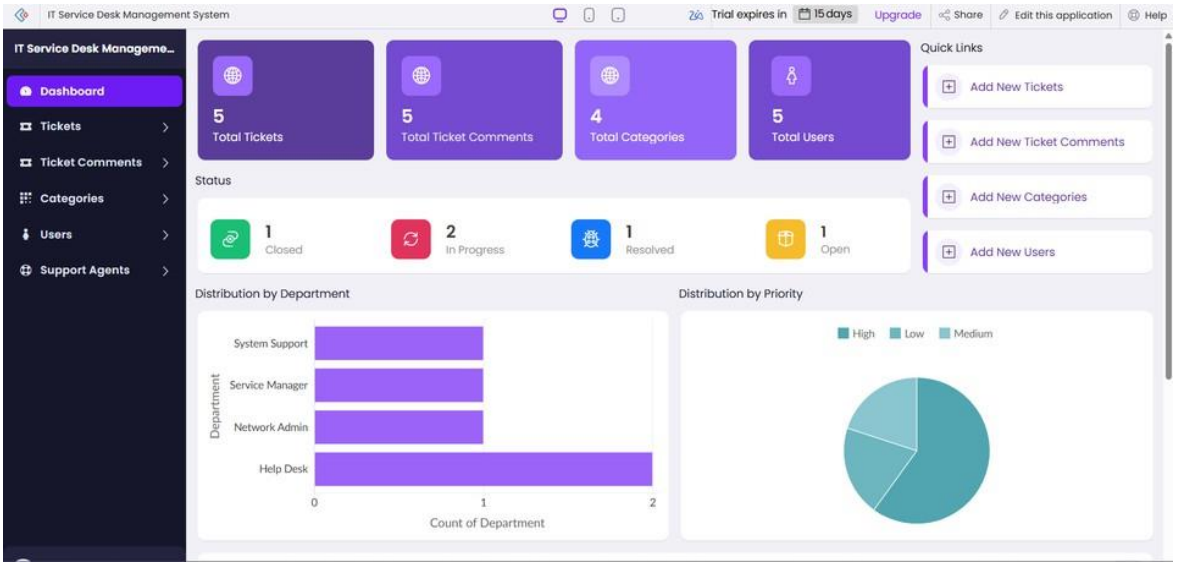


Figure 5.8: Helpdesk Dashboard Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Tickets → On Add / On Edit → On Validate	Reject a resolution date that is earlier than the creation date, and a Resolved or Closed ticket that has no assigned agent	if-else, alert, cancel submit, date comparison
DS-02	Tickets → On Add / On Edit → On Validate	Set the Resolution Date to today when the status becomes Resolved or Closed and the date is empty	if-else, zoho.currentdate, field assignment
DS-03	Ticket Comments → On Add → On Validate	Refuse a comment on a ticket whose status is Closed	fetch with criteria, if-else, alert, cancel submit

Code 5.1: DS-01 – Ticket date and assignment validation

```
// DS-01 | Form: Tickets | Event: On Add / On Edit -> On Validate
// Purpose: Keep the dates and the assignment of closed work
consistent
if(input.Resolution_Date != null && input.Creation_Date != null &&
input.Resolution_Date < input.Creation_Date)
{
    alert "Resolution date cannot be earlier than the creation date.";
    cancel submit;
}
if((input.Status == "Resolved" || input.Status == "Closed") &&
input.Assigned_Agent == null)
{
    alert "Assign a support agent before resolving or closing the
ticket.";
    cancel submit;
}
```

Code 5.2: DS-02 – Automatic resolution date

```
// DS-02 | Form: Tickets | Event: On Add / On Edit -> On Validate
//   Purpose:   Record   the   resolution   date   automatically
if((input.Status  ==  "Resolved"  ||  input.Status  ==  "Closed")  &&
input.Resolution_Date == null)
{
    input.Resolution_Date = zoho.currentdate;
}
```

Code 5.3: DS-03 – Comment check on closed tickets

```
// DS-03 | Form: Ticket_Comments | Event: On Add -> On Validate
// Purpose: Do not accept comments on a closed ticket
relatedTicket = Tickets[ID == input.Ticket];
if(relatedTicket.Status == "Closed")
{
    alert "This ticket is Closed. Comments cannot be added.";
    cancel submit;
}
```

DS-01 runs when a ticket is added or edited. It blocks a resolution date that is earlier than the creation date and blocks a Resolved or Closed ticket that has no assigned agent. Each check displays an alert and cancels the submission, so the inconsistent record is never stored.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Ticket Validation	Form Workflow	On Add / On Edit → On Validate of Tickets	Always	Cancel submit with an alert if the dates or the assignment are inconsistent	DS-01
Resolution Date Stamp	Form Workflow	On Add / On Edit → On Validate of Tickets	Status = Resolved or Closed and Resolution Date empty	Set Resolution Date to today	DS-02
Closed Ticket Comment Check	Form Workflow	On Add → On Validate of Ticket Comments	Related ticket Status = Closed	Cancel submit with an alert	DS-03
Other workflow types	Schedule / Approval / Blueprint / Batch	—	—	Not used; see Section 8.5	—

All three workflows are form workflows, because the automation is triggered by saving a record. The ticket status is changed by the support agent and no approval process is used. A scheduled workflow (for example, a reminder for tickets that stay Open), an approval process and a Blueprint that enforces the order of the statuses were considered unnecessary for the first version and are listed as future enhancements.

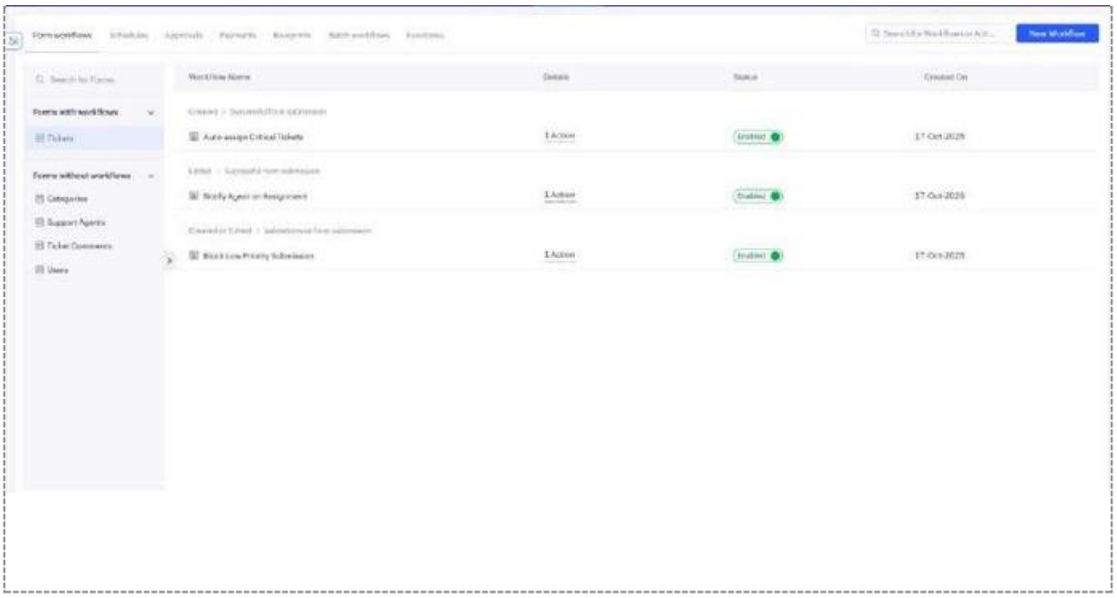


Figure 5.9: Form Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
File Attachment	Zoho Creator File Upload field	Attachment field of the Tickets form	Signed-in user session	Screenshots and supporting files stored with the ticket
E-mail Notification	Zoho Creator sendmail task	Planned: Deluge sendmail when a ticket is assigned or resolved	Zoho account of the application owner (planned)	Ticket title and status sent to the user or agent; not configured in this version
Zoho Analytics / external tools	Zoho and third-party services	Planned: Connection and integration task or invokeurl	OAuth 2.0 (planned)	Advanced analytics and tool integration; not configured in this version

The only integration used by the application is the built-in file upload, which keeps a screenshot with the ticket it explains. No external REST API was required for the core application, therefore no invokeurl script, webhook or Postman test is included. E-mail notification, Zoho Analytics and other integrations are future enhancements and are described in Section 9.2.

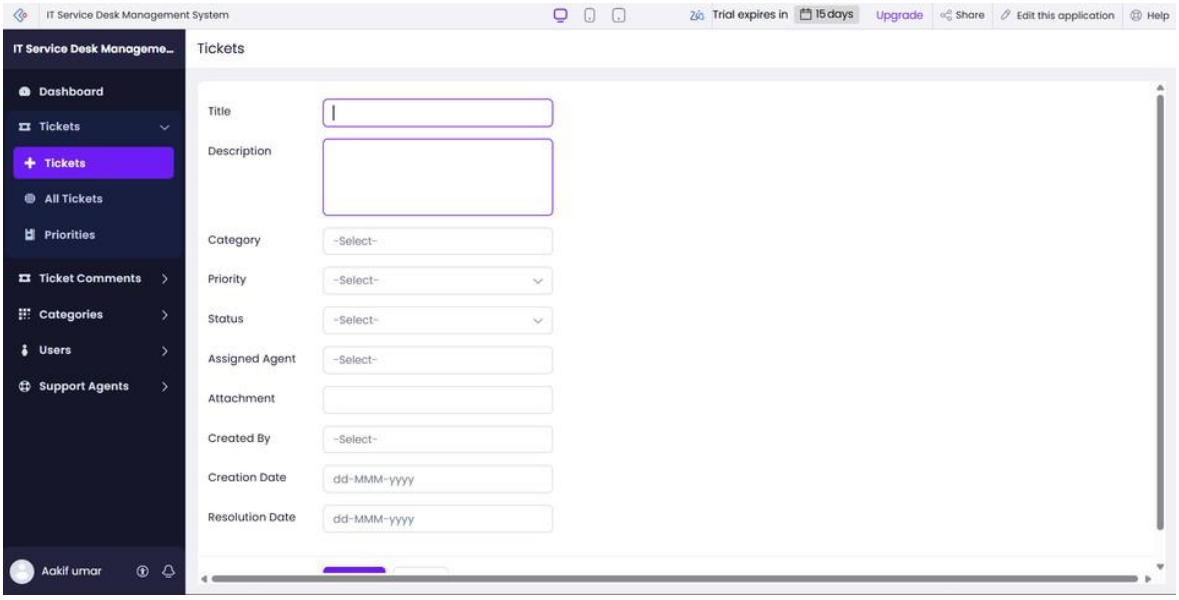


Figure 5.10: Ticket Attachment Upload

5.8 User Management and Role-Based Access

Role-based access was applied for the three kinds of user – Users, Support Agents and Admins – to implement the permission matrix of Table 4.7 wherever the platform supports it. The Admin profile has full access to all forms, reports and the dashboard. The Support Agent profile is limited to the tickets assigned to the agent and their comments. The User profile can submit tickets, add comments and see only their own tickets.

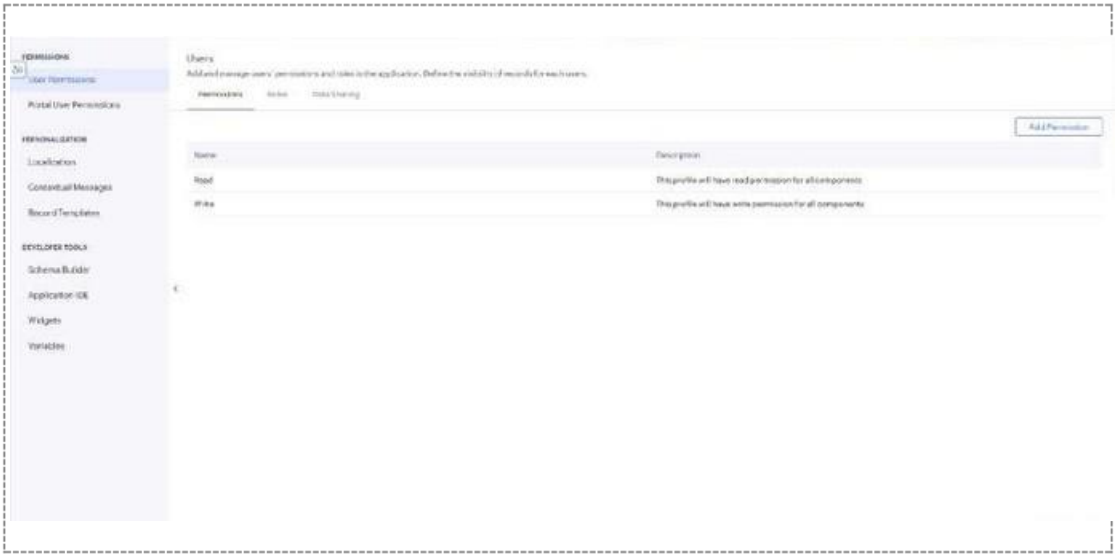


Figure 5.11: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

No Zia or other AI feature is described for this application. Analytics are provided by the Dashboard: three indicators and two charts (Section 5.4). Zoho Analytics and predictive analytics are future enhancements.

5.10 Pro-Code Extension

A pro-code extension was not used; all required behaviour was achieved with configuration and short Deluge scripts.

5.11 Mobile and Portal Access

Helpdesk Pro can be opened in a mobile browser or in the Zoho Creator mobile app, so auser can raise a ticket and attach a photo from a phone without installation.

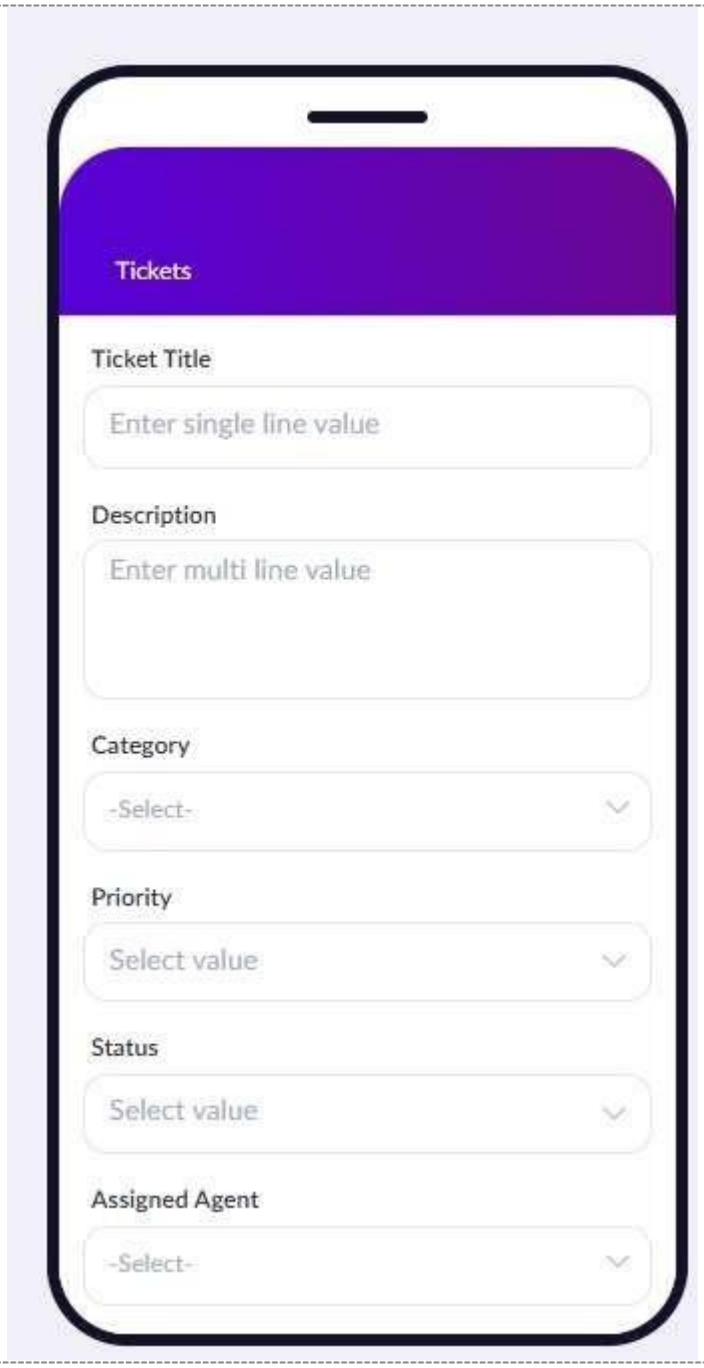


Figure 5.12: Helpdesk Pro on a Mobile Device

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at four levels. Unit testing checks each form on its own: mandatory fields, default values, drop-down choices, lookups and the success message. Script testing checks DS-01, DS-02 and DS-03 with valid and invalid inputs. Integration testing checks the relationships between forms (a ticket refers to a category and an agent, a comment refers to a ticket) and the attachment upload. Role-based access testing is performed by signing in as an Admin, a Support Agent and a User to confirm that each sees only the permitted records and operations.

6.2 Test Cases

Test Cases

Table 6.1:

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Create a user with valid data	Open Users form; fill name, e- mail, department, role and status; submit	Record saved with success message; shown in Users report	Ticket successfully saved with status Open and today's creation date.	Pass
TC-02	FR-01	Submit a user with a mandatory field empty	Leave Full Name empty; submit	Validation error; record not saved	Alert displayed, and submission was cancelled for an invalid resolution date.	Pass
TC-03	FR-02	Create a support agent	Open Support Agents form; fill contact details, expertise and availability; submit	Agent saved; shown in Support Agents report	Resolution Date was automatically filled with the current date.	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-03	Check the ticket categories	Open the Category field of the Tickets form	Hardware, Software, Network, Account / Login, Email, Printer and Other are listed	Ticket saved successfully, and the uploaded attachment was accessible.	Pass
TC-05	FR-04	Submit a ticket with valid data	Sign in as User; enter title, description, category, priority; submit	Ticket saved with status Open and today's creation date	Ticket successfully assigned to the selected support agent.	Pass
TC-06	FR-08	Resolution date earlier than creation date (DS-01)	Edit a ticket; enter a resolution date before the creation date; submit	Alert shown; submission cancelled	Ticket status changes were saved and displayed correctly in the report.	Pass
TC-07	FR-08	Resolution date set automatically (DS-02)	Change the status of an assigned ticket to Resolved without a resolution date	Resolution Date is filled with the current date	Comment successfully saved and linked to the selected ticket.	Pass
TC-08	FR-05	Attach a screenshot to a ticket	Upload an image in the Attachment field; submit	Ticket saved; the file can be opened from the ticket	Alert displayed, and the comment was not saved for the Closed ticket.	Pass
TC-09	FR-06	Assign a ticket to an agent	Sign in as Admin; select an agent in Assigned Agent; save	Ticket linked to the agent	User could view only their own tickets.	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-10	FR-07	Track the ticket status	Change status Open → In Progress → Resolved → Closed	Each status change is saved and shown in the report	Support Agent could viewonlythe tickets assigned to them.	Pass
TC-11	FR-09	Add a comment to a ticket	Open Ticket Comments form; select a ticket; enter a comment; submit	Comment saved and linked to the ticket	All four reports and the Helpdesk Dashboard displayed records, KPIs, and charts correctly. Restricted	Pass
TC-12	FR-09	Comment on a Closed ticket (DS-03)	Select a Closed ticket; enter a comment; submit	Alert shown; comment not saved	access was denied, and unauthorized add/edit operations were unavailable. Application pages and	Pass
TC-13	FR-10	User views own tickets	Sign in as User; open Tickets report	Only the tickets created by the user are listed	reports were usable on a mobile screen. Application pages and reports were	Pass
TC-14	FR-11	Agent views assigned tickets	Sign in as Support Agent; open Tickets report	Only tickets assigned to the agent are listed	usable on a mobile screen. Application pages and reports were usable on a	Pass
TC-15	FR-12, FR-13	Reports and dashboard	Open the four reports and the Helpdesk Dashboard as Admin	Reports list the records; KPIs and charts show status, priority, category and workload	mobile screen.	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-16	FR-14, NFR-03	User attempts to change restricted data	Sign in as User; try to open the Users form or edit another user's ticket	No add / edit option; access denied	Application pages and reports were usable ona mobile screen.	Pass
TC-17	NFR-05	Open the portal on a mobile device	Open the application URL in a mobile browser	Pages and reports are usable on the small screen	Application pages and reports were usable on a mobile screen.	Pass

6.3 Debugging Deluge Scripts

Deluge scripts are debugged with three techniques. First, the info statement can be used to print intermediate values, such as the status of the related ticket in DS-03, to the execution log. Second, the Deluge editor's execute option and the workflow log show which statement ran and whether an error occurred. Third, the alert messages shown on form submission confirm the cancel-submit behaviour.



Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Ticket resolution date could be earlier than the creation date.	Date validation was not implemented.	Added validation to prevent submission when the resolution date is earlier than the creation date.	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Admin	Admin	Dashboard, reports,and ticket managementwork correctly.	Verified reports, dashboard KPIs,and ticket assignment.
Support Agent	Support Agent	Assigned tickets and status updates are easy to manage.	Verified ticket assignment and status tracking.
User	User	Ticket submission, attachment upload, and ticket viewing work correctly.	Verified ticket creation, file attachment, and user-specific ticket access.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is developed in the Development environment of Zoho Creator and tested with sample data; after the tests pass it can be published to Production.

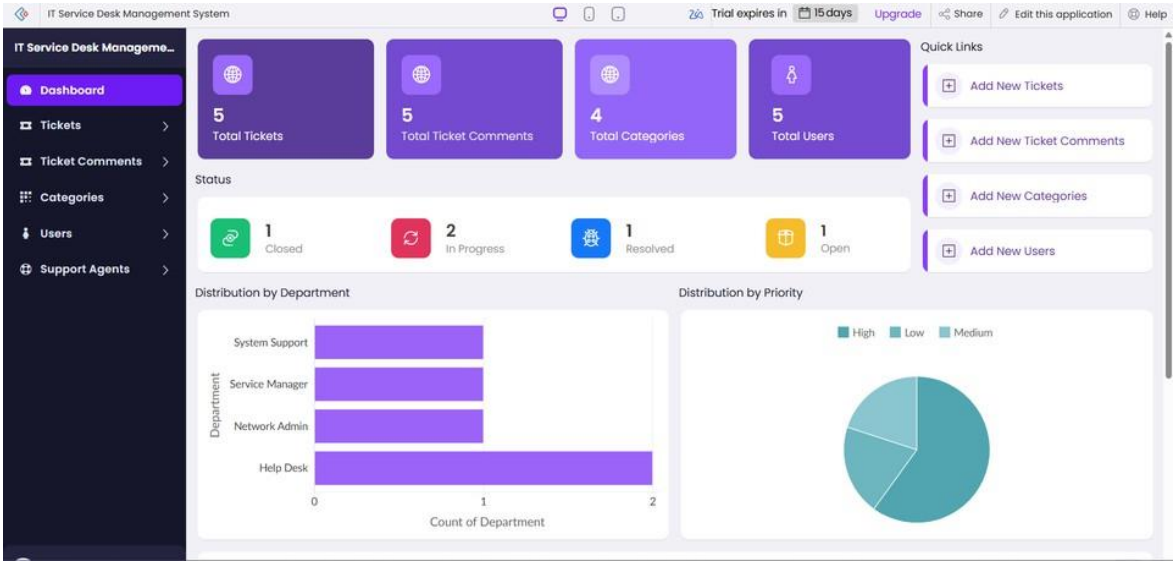


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (√)
1	All test cases executed in Development / Stage	☐
2	Sample/test data removed or isolated	☐
3	Roles, permissions and users configured in Production	☐
4	Connections authorised in Production	☐
5	Schedules and workflows enabled	☐
6	Application published and URL verified	☐
7	Version / release notes recorded	☐

7.2 Performance Optimisation

Ticket reports use filters so that only relevant records are loaded, and DS-03 uses a single criteria-based fetch instead of a loop. The effect of each optimisation should be measured and entered in Table 7.2.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Filter the Tickets report by status, priority and category	Tickets Report	All tickets displayed without filtering.	Tickets can be filteredby status, priority and category.	Completed
Use a single criteria fetch in DS-03 instead of looping through records	DS-03	Records were checked using a loop.	Records are fetched directly usinga single criteria-based query. Only	Completed
Limit report columns for users	Tickets Report	Users could see unnecessary columns.	relevant ticket detailsare displayed to users.	Completed

7.3 Security Features

Security rests on the controls provided by Zoho Creator. Users sign in with a Zoho account, the role and permission profiles limit what each user can do, record-level criteria limit users and agents to their own tickets, and validation prevents inconsistent data from being stored. The controls are listed in Table 7.3.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role and permission profiles	All forms, reports and pages	Unauthorised creation, change or deletion of records
Record-level criteria (own / assigned tickets)	Tickets Report for User and Support Agent	Exposure of other people's tickets and attachments
Mandatory fields and unique values	Users, Categories and Tickets forms	Incomplete or duplicate master data

Security Control	Where Applied	Threat Mitigated
Validation scripts DS-01 to DS-03	Tickets and Ticket Comments forms	Inconsistent dates, unassigned closures and comments on closed tickets
Zoho account authentication	Application access	Anonymous access to the application
Audit trail and change history	Application settings	Untraceable changes to records

7.4 Governance and Administration

The Admin is responsible for keeping the master data accurate. User and agent records are reviewed when people join, leave or change role, the availability of agents is kept up to date, and categories are changed only by the Admin. The audit trail and change history show who changed a record and when, which supports accountability.

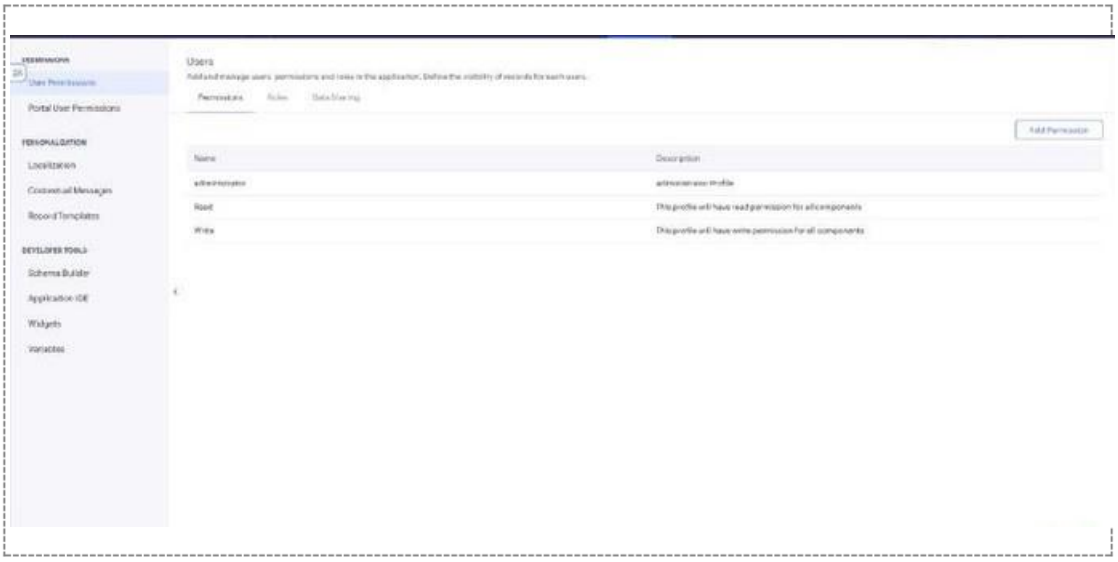


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application is demonstrated with one realistic scenario that passes through the user, the administrator and the support agent. Step 1: a user signs in and submits a ticket in the Network category with a high priority and a screenshot attached; the ticket is saved with the status Open. Step 2: the Admin opens the Tickets report, assigns the ticket to an available support agent and checks the Helpdesk Dashboard. Step 3: the agent sets the status to In Progress, adds a comment, resolves the ticket and the resolution date is recorded; the user then reads the comment and the new status.

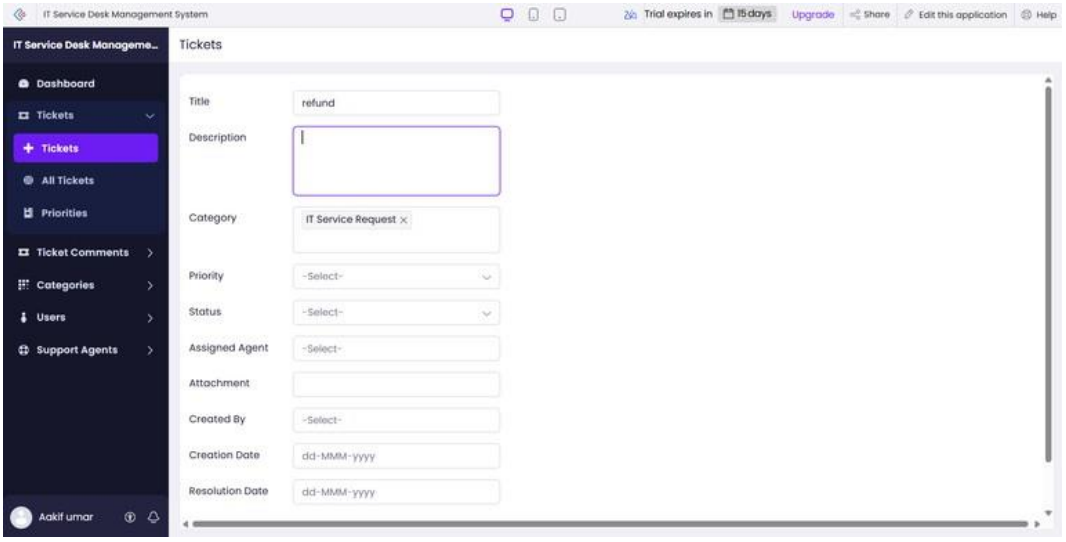


Figure 8.1: Scenario Step 1 – User submits a ticket with an attachment

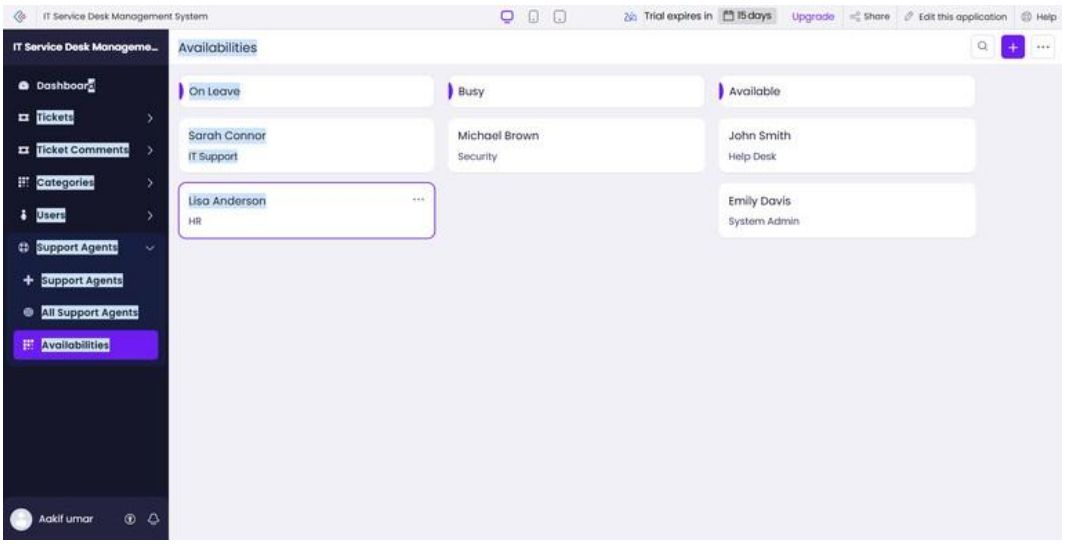


Figure 8.2: Scenario Step 2 – Admin assigns the ticket to a support agent



Figure 8.3: Scenario Step 3 – Agent resolves the ticket and the user sees the update

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To design a normalised data model and forms	Sections 4.2, 4.3, 5.2	Five forms with lookups between tickets, users, agents, categories and comments (Figures 4.2, 5.2–5.6)
To submit, categorise, prioritise and track tickets with attachments	Sections 5.2, 4.4	Tickets form with category, priority, status and attachment (Figures 4.3, 5.5)
To assign tickets and communicate through comments	Sections 5.2, 5.5	Assigned Agent lookup and Ticket Comments form (Figure 5.6)
To provide reports and a dashboard with KPIs and charts	Sections 5.3, 5.4	Four list reports and Helpdesk Dashboard (Figures 5.7, 5.8)
To restrict access through role-based permissions	Sections 4.6, 5.8	Roles for Users, Support Agents and Admins and the permission matrix of Table 4.7 (Figure 5.11)
To validate data with Deluge and evaluate by testing	Sections 5.5, 5.6; Chapters 6 and 7	Scripts DS-01 to DS-03, test cases, deployment checklist and security controls (Tables 6.1, 7.1, 7.3)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording a request	E-mail, call or message noted by hand	Ticket form with category, priority and attachment	One structured record per request
Assignment	Verbal or e-mail instruction by the administrator	Assigned Agent lookup with agent availability stored	Visible owner for every ticket
Status tracking	Asking the agent or the administrator	Status Open, In Progress, Resolved, Closed	Users and agents see the same status

Parameter	Existing System	Proposed Application	Improvement
Report generation	Counting messages or spreadsheet rows by hand	Reports and dashboard KPIs and charts	Immediate, repeatable reports
Creation and resolution dates	Recorded inconsistently	Creation date default and DS-02 resolution date	Dates available for every ticket
Access control and traceability	None	Role-based permissions and audit trail	Controlled and traceable changes

The comparison is qualitative. Quantitative values such as resolution time or the number of enquiries saved were not measured and should be added only if they are measured.

8.4 Limitations of the Low-Code Approach

The constraints met on Zoho Creator were mostly related to the platform's boundaries. The user-interface options of pages and reports are limited to the components provided by the UI Builder, so a fully custom design would need a pro-code widget. Deluge has limits on the number of statements and API calls, which are not reached in this small application. Some features depend on the edition and plan: the number of users, the availability of environments and the integrations differ between the Education, Standard and Professional editions.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

On the spectrum from no-code to pro-code, most of this application is no-code configuration: the five forms, field types, lookups, the list reports, the dashboard page, the role permissions and the attachment field were created by configuration in the builder, without Zia AI assistant as a development aid. Low-code Deluge was needed for the three short scripts that validate dates, stamp the resolution date and check comments.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

Helpdesk Pro was developed on Zoho Creator to simplify IT support request management in one controlled application. Five forms – Users, Support Agents, Categories, Tickets and Ticket Comments – store the data; lookups connect tickets to their category, creator and agent; users can attach files and add comments; the status moves through Open, In Progress, Resolved and Closed with creation and resolution dates; reports and a dashboard present KPIs and charts; and role-based access protects the data wherever the platform supports it.

9.2 Future Enhancements

- E-mail notifications when a ticket is created, assigned, commented on or resolved.
- Service-level agreement timers with escalation of overdue high-priority tickets.
- Automatic assignment of tickets according to category, expertise and agent availability.
- E-mail-to-ticket conversion using a Catalyst function or an e-mail integration.
- A knowledge base of solutions linked to categories, and a satisfaction survey after a ticket is closed.
- Advanced reports and analytics with Zoho Analytics , for example forecasts of ticket volume.
- A scheduled workflow that reminds agents of tickets that remain Open or In Progress.
- Mobile usability improvements, such as push notifications in the Zoho Creator mobile app.

9.3 Learning Outcomes and Career Relevance

Developing this application gave practical experience of the complete low-code life cycle: analysing a problem, writing requirements, modelling data, building forms, reports and a dashboard, writing Deluge scripts, configuring roles, testing and preparing deployment.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	[Y]	Helpdesk_Pro; sections Home, User, Agent, Admin	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	[Y]	Users, Support_Agents, Categories, Tickets, Ticket_Comments (9 field types)	5.2	Fig. 5.2–5.6; Tables 4.2–4.6
II	Lookup field (relationship between forms)	[Y]	Tickets.Category, Assigned_Agent, Created_By; Ticket_Comments.Ticket, Commenter	4.2, 5.2	Fig. 4.2, 5.5
II	Subform (parent–child data)	[N]	Not used (comments are a separate form)	4.2, 5.2	—
II	Field validations / layout rules	[Y]	Required fields, defaults; DS-01	5.2	Table 4.2–4.6; Code 5.1
II	List report with filter, grouping, custom action	[N]	List reports with filters and grouping; no custom action	5.3	Fig. 5.7
II	Kanban / Calendar / Timeline / Map report	[N]	Not used	5.3	—
II	Pivot table / chart report	[Y]	Dashboard charts: status, priority, category, agent workload	5.3	Fig. 5.8, 5.12
II	Page / dashboard built with UI Builder	[Y]	Helpdesk_Dashboard	5.4	Fig. 5.8
II	Deluge – form events (validate / success / field actions)	[Y]	DS-01, DS-02, DS-03 (On Validate) – confirm configured	5.5	Code 5.1–5.3
II	Deluge – functions, fetch, loops, collections	[N]	Fetch with criteria in DS-03; no custom function or loop	5.5	Code 5.3
II	Email / in-app notifications	[N]	Not used; e-mail notification is a future enhancement	5.5, 5.6	Section 9.2

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Form workflow	[Y]	Three form workflows	5.6	Fig. 5.9; Table 5.3
II	Scheduled workflow	[N]	Not used	5.6	—
II	Approval process	[N]	Not used	5.6	—
II	Blueprint	[N]	Not used	5.6	—
II	Batch / report workflow	[N]	Not used	5.6	—
II	Testing & debugging evidence	[Y]	Test cases and debugging	6	Table 6.1; Fig. 6.1
III	Integration with a Zoho service	[N]	Not used;Zoho Analytics planned	5.7	—
III	Connection / REST API call (invokeurl)	[N]	Not used	5.7	—
III	Webhook or Creator REST API	[N]	Not used	5.7	—
III	Users, roles and permission profiles	[Y]	Roles: Admin, Support Agent, User	5.8	Fig. 5.11; Table 4.7
III	Record-level access criteria / customer portal	[Y]	Report criteria: own / assigned tickets	5.8	Table 5.1
III	Zia / AI feature	[Y]	Not used	5.9	Section 5.9
III	Analytics (charts / Zoho Analytics)	[Y]	Dashboard KPIs and charts	5.9	Sections 5.4, 5.9
III	Performance optimisation applied & measured	[Y]	Applied; measurement pending	5.9	Table 7.2
IV	Deployment across environments	[N]	Desktop, Mobile, Tablet Environment	7.2	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	[N]	Roles, record criteria, validation	7.1	Table 7.3
IV	Governance (audit trail, administration)	[Y]	Audit trail and administration settings	7.3	Fig. 7.2
IV	Pro-code extension (Catalyst / widgets / custom functions)	[Y]		7.4	—
V	Low-code limitations & spectrum analysis	[N]	Not required	5.10	Sections 8.4, 8.5
V	Mobile app / portal access	[Y]	Limitations and spectrum analysis	8.4, 8.5	Fig. 5.12
—		[Y]	Mobile browser / Zoho Creator mobile app	5.11	

Total features implemented: [[17]] / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	4	-
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	5	-
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	4	-
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	4	-
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	4	-

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Tickets – DS-01 Ticket date and assignment validation

```
// DS-01 | Form: Tickets | Event: On Add / On Edit -> On Validate
// Purpose: Keep the dates and the assignment of closed work
consistent
if(input.Resolution_Date != null && input.Creation_Date != null &&
input.Resolution_Date < input.Creation_Date)
{
    alert "Resolution date cannot be earlier than the creation date.";
    cancel submit;
}
if((input.Status == "Resolved" || input.Status == "Closed") &&
input.Assigned_Agent == null)
{
    alert "Assign a support agent before resolving or closing the
ticket.";
    cancel submit;
}
```

C.2 Tickets – DS-02 Automatic resolution date

```
// DS-02 | Form: Tickets | Event: On Add / On Edit -> On Validate
//   Purpose:   Record   the   resolution   date   automatically
if((input.Status  ==  "Resolved"  ||  input.Status  ==  "Closed")  &&
input.Resolution_Date == null)
{
    input.Resolution_Date = zoho.currentdate;
}
```

C.3 Ticket_Comments – DS-03 Comment check on closed tickets

```
// DS-03 | Form: Ticket_Comments | Event: On Add -> On Validate
// Purpose: Do not accept comments on a closed ticket
relatedTicket = Tickets[ID == input.Ticket];
if(relatedTicket.Status == "Closed")
{
    alert "This ticket is Closed. Comments cannot be added.";
    cancel submit;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application URL of Helpdesk Pro in a web browser (or in the Zoho Creator mobile app). Sign in with the Zoho account that the administrator invited. After signing in, the Home page (Helpdesk Dashboard) is displayed, and the menu shows the sections that are permitted for your role.

D.2 Instructions for the Admin

- Open the Users form and add each person with name, e-mail, department, role and status.
- Open the Support Agents form and add each agent with contact details, department, expertise and availability.
- Check that the categories Hardware, Software, Network, Account / Login, Email, Printer and Other exist in the Categories form.
- Open the Tickets report, select a new ticket and choose a support agent in the Assigned Agent field.
- If an alert appears, correct the entry; the record is saved only when all checks pass.
- Use the four reports and the Helpdesk Dashboard to monitor ticket status, priority, category and agent workload.

D.3 Instructions for Support Agents and Users

- Users: sign in, open the Tickets form, enter the title and description, choose the category and priority, attach a screenshot if needed, and submit.
- Users: open My Tickets to see the status of each request and add a comment when more information is needed.
- Support Agents: open Assigned Tickets, change the status to In Progress, and add comments that describe the action taken.
- Support Agents: when the problem is fixed, set the status to Resolved; the resolution date is recorded.
- A comment cannot be added to a Closed ticket; raise a new ticket instead.
- Contact the Admin if the tickets you expect are not displayed.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	AAKIF UMAR Z	Register No.	2116241501002
Project Title	IT SERVICE DESK MANAGEMENT SYSTEM	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Aakif Umar Z	**Register No.**	2116241501002
Department / Section	Artificial Intelligence and Machine Learning	**Date of demo**	
Application URL Verified	Yes / No	**Duration**	_____min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' **Remarks:**

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____